The purpose of this document is to provide steps to run EDGIE.

EDGIE is a MATLAB based tool to analyze the impacts of electric vehicles, heat pumps, and heat-pump water heaters on the distribution grid. It also includes a vehicle-to-home strategy to reduce the peak demand.

To run the V2H formulation it will require CVX and MOSEK to be installed.

The authors used MATLAB 2022, CVX 2.2 and Tableau 2023.3 to generate and plot the results.

The next page will include steps for downloading the required input files and running the EDGIE code.

It will require 35 GB of space.

Step 1: Create a folder named "ComStock"

Step 2: Go to the link : 10.5281/zenodo.21628313

Step 3: Download the three zip files

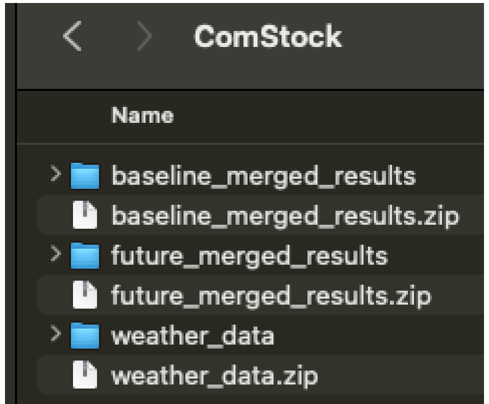
Files (12.6 GB)

Name	Size	
baseline_merged_results.zip md5:42bb7cba9b1c8f5f04642c6038e7ee18	6.0 GB	Preview Download
future_merged_results.zip md5:c2635850c609ca0032424e755e2a5f43	6.0 GB	Preview Download
weather_data.zip md5:cd3f6ca9e79e6e533bdaf1a491b08000	539.7 MB	Preview Download

Step 4: Move the downloaded files to "ComStock" folder

Step 5: Extract all the files

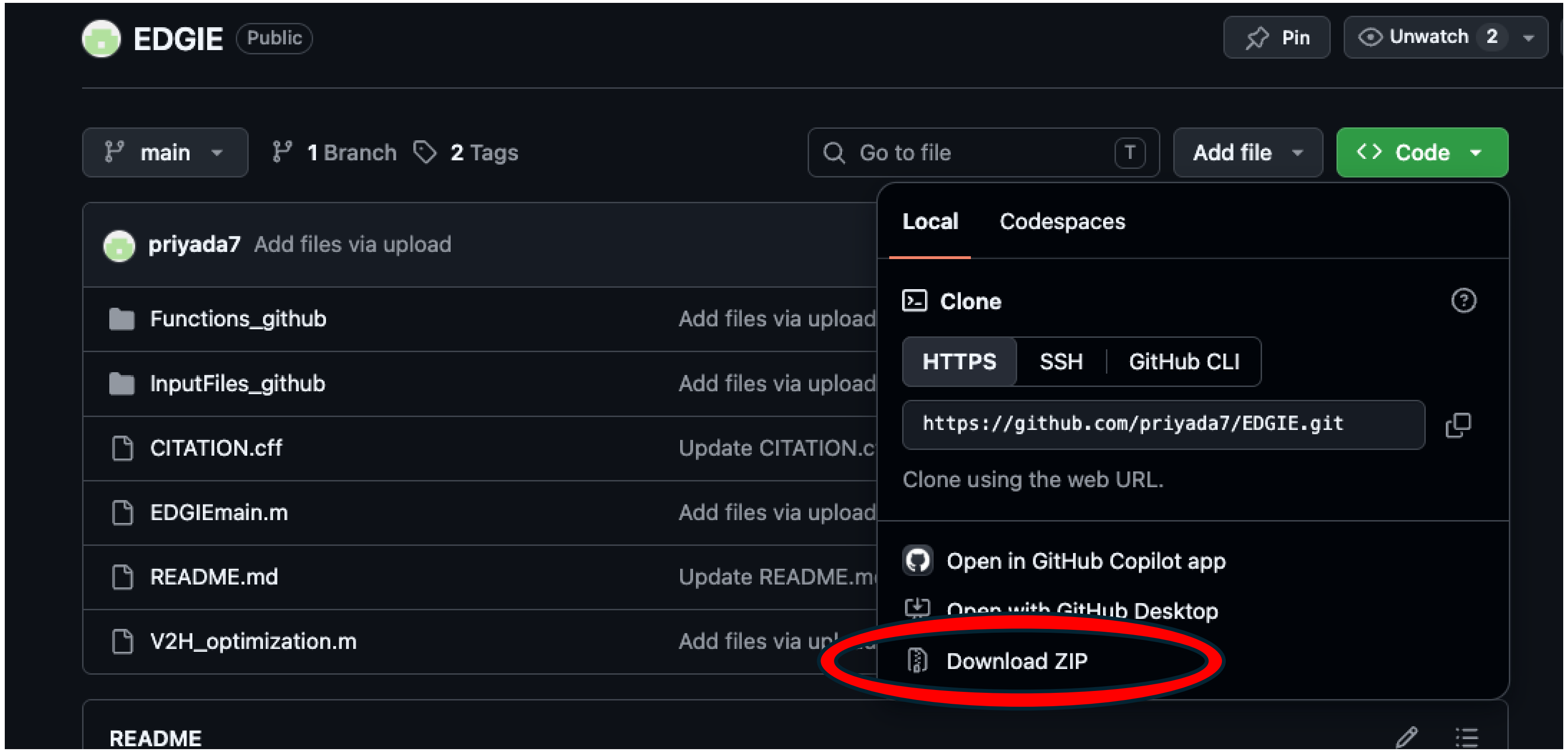
Step 6: ComStock folder should look like this



Step 7: Go to the github repository:

https://github.com/priyada7/EDGIE

Step 8: Click on code and then select download zip



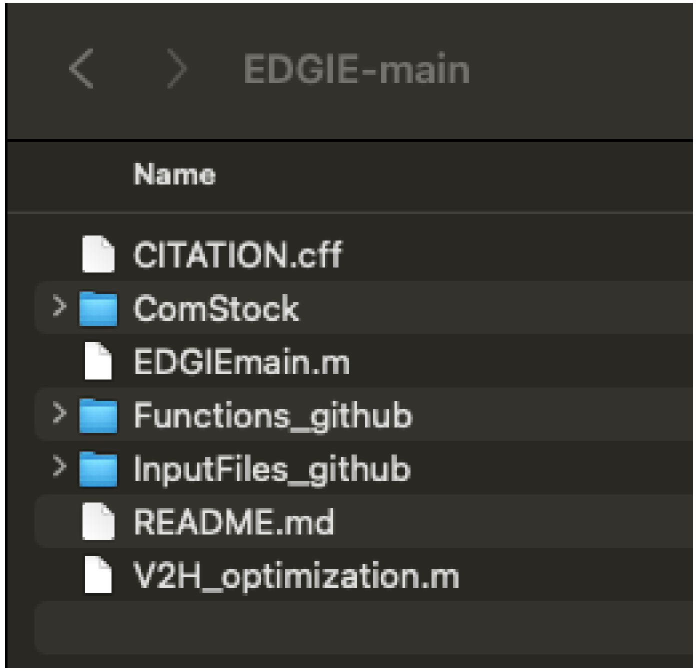
Step 9: Create a new folder “EDGIE Github”

Step 10: Move the zip folder EDGIE-main.zip in that folder

Step 11: Extract the zip file

Step 12: Move the ComStock folder to “EDGIE-main” folder

Step 13: The “EDGIE-main” should look like this



Step 14: Open EDGIEmain.m file

Step 15: Replace XXX with the working directory location (line 2- 5)

```
addpath 'XXX/EDGIE Github/EDGIE-main/Functions_github'
addpath 'XXX/EDGIE Github/EDGIE-main/InputFiles_github'

baseDir = 'XXX/EDGIE Github/EDGIE-main/ComStock';
```

Example : for macOS

```
addpath '/Users/priyadarshan/Library/CloudStorage/Box-Box/Herricks/Codes/EV/v2h/v2h new/untitled folder/EDGIE Github/EDGIE-main/Functions_github'
addpath '/Users/priyadarshan/Library/CloudStorage/Box-Box/Herricks/Codes/EV/v2h/v2h new/untitled folder/EDGIE Github/EDGIE-main/InputFiles_github'

baseDir = '/Users/priyadarshan/Library/CloudStorage/Box-Box/Herricks/Codes/EV/v2h/v2h new/untitled folder/EDGIE Github/EDGIE-main/ComStock';
```

Step 16: Click Run

By default it will run for Minnesota and Hennepin county (stated = 27) and output in command window should be of the order 5e9.

Note: The exact number may vary depending on the random numbers generated, version of MATLAB.

To run statewide location, change line 171 to

```
for stateIdx = 1:length(cities)
```

To run the V2H code,

It will first require installing CVX 2.2.

CVX can be downloaded from : https://cvxr.com/cvx/

Open the file V2H_optimization.m and specify the working directory and click run

Note: optimization will be slow, it is advised to use MOSEK. One location simulation should be complete in 40-50 seconds.

Software requirement:

MATLAB

https://www.mathworks.com/help/install/ug/install-products-with-internet-connection.html

CVX

https://cvxr.com/cvx/

Plotting results

https://www.tableau.com/